# CFD 2025-26 – Lab 08

## Exercise 1

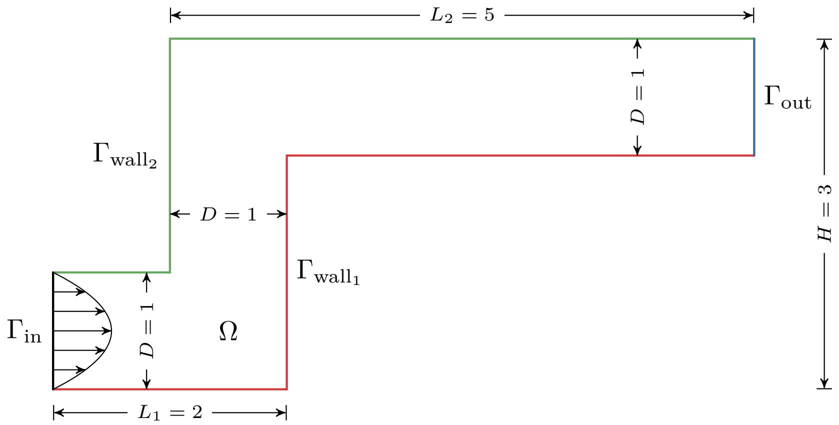


Figure 1: Domain $\Omega$ for Exercise 1.

Let us consider the stationary Navier-Stokes problem defined in the domain $\Omega$ as in Figure **??** (defined in `elbow3.msh` and `elbow3bis.msh`):

$$\begin{cases} (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} - \operatorname{div}(2\nu\varepsilon(\boldsymbol{u})) + \nabla p = \boldsymbol{0}, & \text{in } \Omega, \\ \operatorname{div} \boldsymbol{u} = 0, & \text{in } \Omega, \end{cases}$$

where the kinematic viscosity $\nu$ is equal to $2 \cdot 10^{-2}$, the strain tensor is $\varepsilon(\boldsymbol{u}) = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T)$, and the boundary conditions are:

- a parabolic profile of velocity $\boldsymbol{u}_{\text{in}} = 4y(1-y)\boldsymbol{i}$ on the inlet boundary $\Gamma_{\text{in}}$;
- stress-free conditions on the outlet boundary $\Gamma_{\text{out}}$.
- one of the following on the lateral walls $\Gamma_{\text{wall}_1} \cup \Gamma_{\text{wall}_2}$:
  - (i) no-slip conditions: $\boldsymbol{u} = \boldsymbol{0}$;
  - (ii) free-slip conditions: $\boldsymbol{u} \cdot \boldsymbol{n} = 0, \quad (2\nu\varepsilon(\boldsymbol{u}) - pI)\boldsymbol{n} \cdot \boldsymbol{t} = 0$, where $\boldsymbol{t} \perp \boldsymbol{n}$.

1. Derive the weak formulations of the nonlinear problems corresponding to both case (i) and (ii), using a strong imposition of Dirichlet conditions.

2. Consider the no-slip case (i).
Starting from the provided template, implement the problem in Firedrake, using its residual-based nonlinear solver and export to Paraview the velocity $\boldsymbol{u}$, pressure $p$ and

stream function $\psi$ defined as the solution of the following elliptic boundary value problem

$$\begin{cases} -\Delta\psi = \omega, & \text{in } \Omega, \\ \nabla\,\psi \cdot \boldsymbol{n} = 0, & \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \\ \psi = 2/3, & \text{on } \Gamma_{\text{wall}_1}, \\ \psi = 0, & \text{on } \Gamma_{\text{wall}_2}, \end{cases}$$

where $\omega$ is intensity of the vorticity field. Use $\mathbb{P}^1$ finite elements for $\psi$.
**<u>NB</u>**: in 3 dimensions, $\omega = |\boldsymbol{\omega}| = |\nabla \times \boldsymbol{u}|$. For 2-dimensional flows, the vorticity field is defined as orthogonal to the flow plane, with intensity $\omega = \partial_x u_y - \partial_y u_x$.

3 Compute the following quantities:

- drag force exerted by the fluid onto the lower downstream wall $\Gamma = \{x \in (2, 6), y = 2\}$

$$F_{\text{D}} = -\int_{\Gamma} (2\nu\varepsilon(\boldsymbol{u}) - pI)\boldsymbol{n} \cdot \boldsymbol{i}\, ds;$$

- flowrate through the outlet $\Gamma_{\text{out}}$

$$Q_{\text{out}} = \int_{\Gamma} \boldsymbol{u} \cdot \boldsymbol{n}\, ds;$$

- total pressure jump

$$\Delta p = p_{\text{in}} - p_{\text{out}} = \frac{1}{|\Gamma_{\text{in}}|}\int_{\Gamma_{\text{in}}} p\, ds - \frac{1}{|\Gamma_{\text{out}}|}\int_{\Gamma_{\text{out}}} p\, ds.$$

4 Modify the code to solve case (ii) and compare the results both in Paraview and in terms of $F_{\text{D}}, \Delta p$: what can you observe?

## Exercise 2 - homework

Consider the following time-dependent, nonlinear Navier-Stokes problem:

$$\begin{cases} \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} - \frac{2}{\text{Re}}\,\text{div}(\varepsilon(\boldsymbol{u})) + \nabla p = \boldsymbol{0} & \text{in } \Omega, t \in (0, T), \\ \text{div}\,\boldsymbol{u} = 0 & \text{in } \Omega, t \in (0, T), \\ \boldsymbol{u} = 1\boldsymbol{i} & \text{on } \Gamma_{\text{in}} = \{\boldsymbol{x} \in \partial\Omega, x = -3\}, t \in (0, T), \\ \left(\frac{2}{\text{Re}}\varepsilon(\boldsymbol{u}) - pI\right)\boldsymbol{n} = \boldsymbol{0} & \text{on } \Gamma_{\text{out}} = \{\boldsymbol{x} \in \partial\Omega, x = 12\}, t \in (0, T), \\ \boldsymbol{u}\cdot\boldsymbol{n} = 0, \quad \left(\frac{2}{\text{Re}}\varepsilon(\boldsymbol{u}) - pI\right)\boldsymbol{n}\cdot\boldsymbol{t} = 0 & \text{on } \Gamma_{\text{wall}} = \{\boldsymbol{x} \in \partial\Omega, y = -2, 2\}, t \in (0, T), \\ \boldsymbol{u} = \boldsymbol{0} & \text{on } \Gamma_{\text{cyl}} = \{\boldsymbol{x} \in \partial\Omega, (2x)^2 + (2y)^2 = 1\}, t \in (0, T), \\ \boldsymbol{u} = \boldsymbol{0} & \text{in } \Omega, t = 0, \end{cases} \tag{1}$$

where the domain $\Omega$ is provided in `cylinder-ns.msh`, $T = 10$, $\text{Re} = 10$.

Starting from the code implemented in Exercise 1, and using the hints provided in the template, solve problem (**??**) by the Implicit Euler method with an implicit treatment of the advection term, using $\Delta t = 0.2$ and the Firedrake's nonlinear solver for each time step.

*Warning*: running the solution of the problem may take several seconds for each timestep.